

generate feature values for each token are only then needed and presumed to more or less accurately represent the features encoded by each token's PLC.

By contrast, the German and to an even greater extent the Chinese data presumably encode feature nuances that are not present in the English data, and perhaps even unrepresented in the feature set itself. Because German is relatively close to English both genetically and typologically, feature recovery was easier, which likely explains its intermediate performance.

Chinese, on the other hand, is wholly unrelated to English (or German), and its TAM system operates on a logic which is entirely distinct from what is seen in Indo-European languages generally, to say nothing of Low Germanic

8

languages specifically. Thus it is unsurprising that not only is the token count significantly lower for Chinese, at 115 and 121, but the models' performance on these data is considerably worse, nearing 50%.

Subsequent experiments will seek to counteract anchor language effects through a combination of reserve lookup extraction (i.e. searching for potential PLCs in each language, not just English), recruitment of L1 speaker annotators, and refinement of the feature set.

Feature Landscape Effects

As mentioned in the previous section, with respect to the Chinese TAM system, it generally goes without saying that each language's TAM (in fact, entire grammatical system) operates on its own logic, including what features it chooses to contrast or mark overtly versus those it doesn't, the number of TAM constructions in the system's inventory, and more. Because the bulk of the research conducted on PLCs to date has focused on Indo-European, and especially Germanic languages (English alone represents a significant focus of PLC research historically), the feature set that was developed for the pilot experiment, reflecting research landscape, tends to better reflect the types of features that those languages "care about". For example, if the English perfect is considered a "complex tense", i.e. in the Reichenbachian (1947) sense of a grammatical form which combines tense and aspect, or which divides its event and reference times, to what extent can we truly apply the logic developed for English to a language which, like Chinese, has been called "tenseless" (Lin 2006 and 2010, Collart 2021)?

Tackling this quandary is one of the central aims of this project, by way of developing a feature set which can be used effectively on cross-linguistic and typologically varied languages.

In the next section I present the results by language of the pilot experiment, followed by some related literature



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7:36 PM Sep 15

I would be skeptical of this "because" until I see evidence of a greater isomorphic between constructions and features in English on the one hand and in German on the other (compared to in Chinese).



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7:36 PM Sep 15

Say more



Barend Beekhuizen

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Why? This seems to assume more distinct system leads to more NAs but why would that be?



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7:36 PM Sep 15

Not sure how this would solve the issue (instead of distributing it more equitably across languages maybe?)



Barend Beekhuizen

7:36 PM Sep 15

Similarly not so convincing without elaboration



Barend Beekhuizen

7:36 PM Sep 15

Also, the Dahl and Wälchli paper on iamatives has a lot to say on this matter



used effectively on cross-linguistic and typologically varied languages.

In the next section I present the results by language along with some discussion, followed by some global/cross-linguistic performance discussion.

Results and Interpretations by Language

Below I present the performance of the models by language broken into two categories: error types and feature-value associations. The error type category describes the occurrence patterns by specific error type, such as “predicted *le* → actual *unmarked*”. The feature value associations incorporate a variety of statistical methodologies to better inform analysis of error patterns **by incorporating feature values**. In this section I present the relevant statistic information including feature error rates, *p*-values, ϕ -values, and weighted scores (informing overall feature-based performance). In each language section the quantitative results are presented first (by category), followed by interpretations and discussion of those results. A subsequent section will take all of these data and interpretation collectively to pose global interpretations and discussion.

With respect to the X^2 analyses, the following calculations were used to generate the values presented here:

- **X^2 *p*-values**: the *p*-values were calculated along a 2x2 axis comparing positive and negative occurrence of features and errors; i.e. *f+* *e+* represents the number of tokens that bear a feature and were predicted erroneously, while *f-* *e+* represents the number of tokens that do not bear a feature and were predicted erroneously. This analysis incorporated the Yates correction, which is designed to correct for smaller sample sizes to avoid overstating significance values.
- **ϕ -values**: the ϕ -values were calculated using the square root of the value of the X^2 -score divided by the total sample size for that table.

⁷ This is my own term, but I’m quite fond of it. I am unsure to what extent it is used elsewhere in the literature; I have seen it, but not a lot.

9

- **Weighted scores (WS)**: these encompass a range of calculations:
 - Baseline error rate (BER): total errors / total tokens (the “expected” error rate if errors were evenly distributed across features)
 - Feature-specific error rate (FER): errors that bear the feature / total occurrences of the feature in the data set



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This is the classic etic vs emic distinction.



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Unclear



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Exemplify and discuss what the purpose is



Barend Beekhuizen

7:36 PM Sep 15

I see now that it’s about finding the association with the presence/absence of an error (are tests one-tailed or two-tailed?). I thought we were looking at feature-by-construction associations



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Unfamiliar to me: what does it do and who suggests to use it?



Barend Beekhuizen

7:36 PM Sep 15

These are over individual features, right? Explicate

